

Submission Summary

Conference Name	IEEE Chandigarh Subsection International Conference
Track Name	Next-Generation Computing and Applications
Paper ID	1295
Paper Title	Comparative Analysis of Sorting Algorithms
Abstract	The activity of sorting is one of the basic computational procedures that can help optimize database queries, retrieve information, schedule tasks for operating systems, design compilers and perform scientific computations. This paper presents a thorough comparative study on six common sorting algorithms: Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort and Heap Sort. The study involves analysis by formal mathematical complexity and by empirical benchmarking on different dataset sizes and structures. The study also has special consideration for partially sorted datasets using the inversion counting theory, a comparison of iterative vs recursive implementations, the guarantees that sorting algorithms provide with respect to stability, how cache-friendly each algorithm is when considering existing memory hierarchies and how much auxiliary memory each algorithm requires. All asymptotic bounds are supported by mathematical proofs using decision tree theory, Master Theorem applications and by being able to solve exact recurrences. Pseudocode for all six algorithms is also provided to facilitate reproduction. The empirical findings of the study are visually represented in two empirical charts: a line graph showing the divergent performance of $\Theta(n \log n)$ algorithms at scale and a grouped bar graph illustrating the adaptivity of the different algorithms regarding input types. The study will also discuss five real-world software engineering contexts to exemplify practical uses of sorting and

perform a complete analysis of production hybrid sorting strategies, including Timsort, Introsort and pdqsort, that integrate insights from multiple sorting algorithms into an effectively utilized sorting strategy across a wide variety of workloads. The overall result of the study provides ten concrete evidence-based principles for algorithm selection for use in practice.

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